

Description

Students will learn more using Construct 3, where they'll make platformer games. They'll practice making simple levels and get really good at creating animation frames with sprite sheets.

Objectives

Students will:

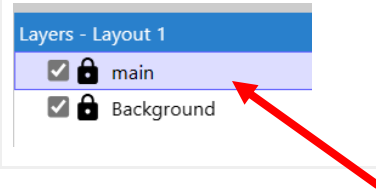
1. Recall the concept and elements of 2D Games.
2. Comprehend type of games : Platformer.
3. Comprehend the concept of sprite sheets and grasp the essence of animation frames.
4. Develop their own design.
5. Build a platformer game.

Tools

1. Construct 3

Class Structure

Duration	Topic
5 mins	Introduction
10 mins	Platformer Game
5 mins	Set the viewport size
5 mins	Adding background
15 mins	Sprite Sheet
15 mins	Player & Candy
10 mins	Adding tileset
15 mins	Adding portal
5 mins	Adding keyboard and text
5 mins	Wrap - up

Planned Activities	Resources
Introduction(5 mins)	
Review : - Teacher asks students what they have learned from their previous module. - Teacher asks students if they still remember what they learned. - Tell the students, in this module they will learn how to create project platformer	
Platformer Game (5 Mins)	
Show & Explain : - Show the students the game we're planning to make. Ask them if they've played platformer games before and if they can name one. - Explain what a platformer game is: it's a game where you control a character to jump and move around platforms to reach goals.	Example : Stickman - Platformer SuperRobo - Platformer Definition of Platformer Game What is Platformer?
Set the viewport size (5 Mins)	
Show & Explain : Give instructions to create a new project and set the layout as well as viewport.	 The screenshot shows a 'Layers - Layout 1' panel with two entries: 'main' and 'Background'. Both have a checked checkbox and a lock icon. A red arrow points to the 'main' entry.

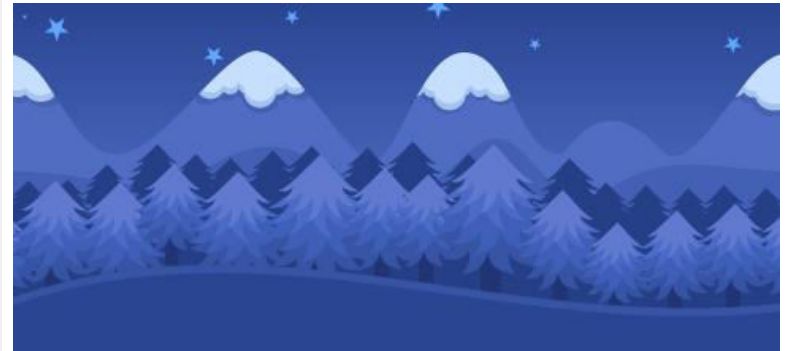
- Set the layout : **1200 x 600**.
- Set the Viewport : **600 x 600**.
- Setting Orientation : **Landscape** .

Display ▼	
Viewport size ▼	600 x 600
Width	600
Height	600
Aspect ratio	1:1

Adding background

Show & Explain :

- Tell the students to add the background picture and make it the right size for the whole layout. Also, have them rename the sprite. Once they're done with the player sprite, they can lock the layer so it won't move by mistake.



Sprite Sheet (15 Mins)

Show & Explain :

- Provide students with explanation sprite sheets and animation frames :
Sprite sheet : A sprite sheet is like a single picture that has lots of different animation frames or game visuals all in one place. Instead of using separate pictures for each frame or visual, sprite sheets put them

Explanation :

[Sprite - Construct 3 Documentation](#)

Video Explanation :

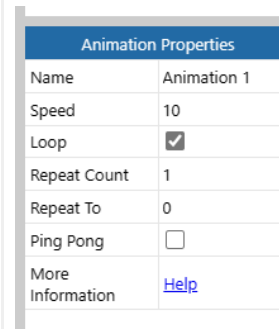
[Tutorial Sprite Sheet](#)

all together in one image. This can make games run faster and use less memory.

Player & Candy (15 Mins)

Show & Explain :

- Let the students know that their next task is to create a new layer and add the sprite just like they usually do.
- Show the students how to add the sprite sheet and then break it into separate frames.
- Tell them to set up the collision settings and choose the image point.



Link of player assets :

<https://www.sprites-resource.com/mobile/cookierun/>

Adding tileset

Show & Explain :

- Tell the students they'll be creating a tile sprite with lots of pictures. .
- Show them how to do it and explain the steps..
- Make sure they understand that they need to split the tileset into parts and set up things like collisions and image points.

Documentation Link :

Animation 1 Frames (11)



0 1 2 3 4 5 6 7

Animation Properties

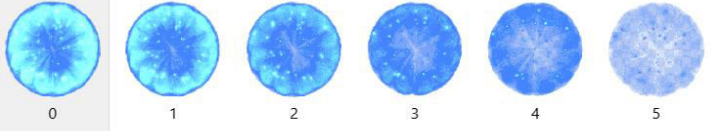
Name	Animation 1
Speed	0
Loop	☐
Repeat Count	1
Repeat To	0
Ping Pong	☐
More Information	Help

Adding portal

Show & Explain :

- Show the students how to add a portal that they need to use to reach the finish line.
 - Tell them that the way they create things stays the same, and they should start by setting up the animation
 - Help them make and add the sprites, then choose the right animation frame.

Animation 1 Frames (6)



0 1 2 3 4 5

Adding keyboard and text

Show & Explain :

- Explain why it's important to have keyboard controls for the player to move in the game. Also, give instructions on how to show the number of candies collected with text on the screen..

 Keyboard

 Text

Individual Assignment:

Showcase how to execute the project to check if the sprites and background are properly set up.

Wrap Up (5 mins)

Wrapping Up

- We wrap up by asking questions to confirm students' grasp of the main concepts. Raise your hands if you can explain what comparison operators and logical operators are.
- We express our gratitude to the students for their commendable effort throughout the session.

☐☐ Good Luck With The Class !! ☐☐